

Assignment: 01

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I

MCQ

- | | |
|-------|-------|
| 1. A | 11. A |
| 2. C | 12. C |
| 3. B | 13. C |
| 4. A | 14. D |
| 5. B | 15. A |
| 6. A | 16. B |
| 7. A | 17. D |
| 8. C | 18. B |
| 9. B | 19. B |
| 10. A | 20. A |

II Descriptive.

1. Given a c unit linear classifiers with corresponding weights w_j $j \in \{1, 2, \dots, c\}$ and discriminant function $g_j(x) = w_j^T(x)$.

Now remove one of the units k from c units classifiers, such that the reconstructed linear machine makes same classification decisions.

- i) The classification decisions are based on the maximum of all the discrimination function value of all classes

$$\hat{y} = \arg \max_j g_j(x) = \arg \max_j (w_j^T x)$$

it comes to 2 cases:

- ii) 1st case: where $g_k(x)$ is never maximum i.e., k is never predicted. then removing k

will never effect the classifies and predicts same classes as before.

ii) 2nd Case :- Where k comes as highest sometimes and input is predicted as class k . To make sure that the classification is not effected by the removal of k , we should adjust the weight vectors of remaining units such that classification remains same. i.e., redistribute the w_k to remaining units.

$$w_j' = w_j + \alpha w_k \quad \forall j \neq k$$

α is small factor that redistributes the w_k to remaining classes. ensures ranking of discriminant is same.

These changes make sure that for every α , that the classifier behaves in the same way as before.

thus helping in maintaining original decision boundaries.

v) Even for the previous k prediction samples will select the next highest after k , while remaining has the same predictions as before as there is equivalent distribution of weight w_k among them. while maintaining consistency.

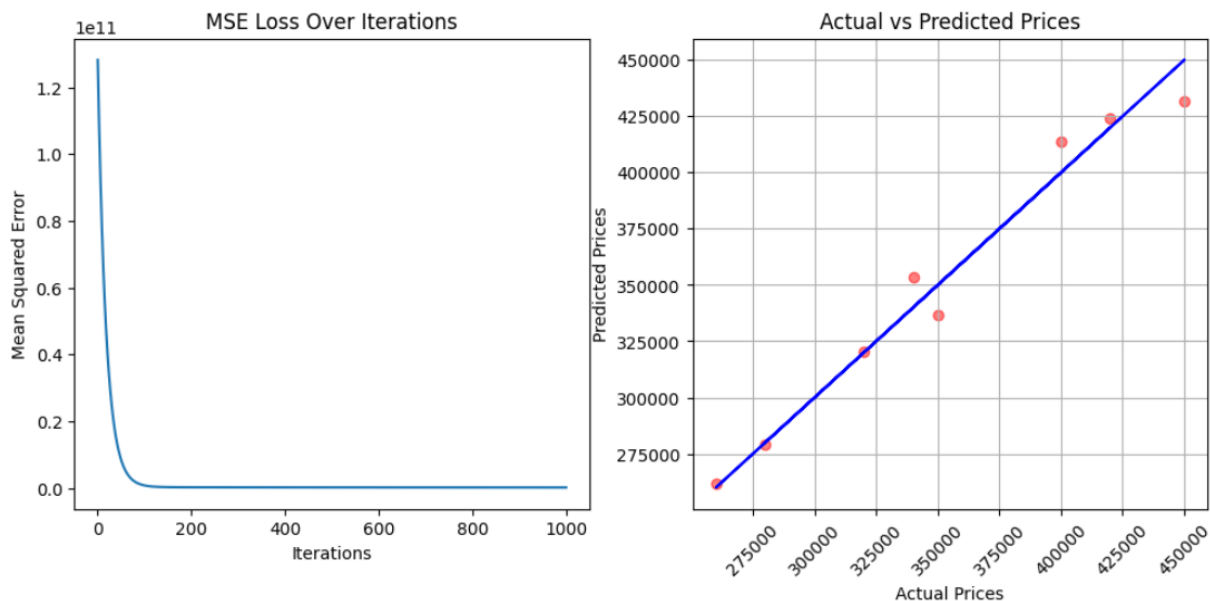
CSCE5210- FUNDAMENTALS OF AI HOMEWORK -01: PART-III CODING QUESTION

PDF REPORT:

1. Implementations and Methodology:

I have built the multiple linear regression for estimating the house price and using the gradient descent to minimize the loss function which in our case is MSE as given.

1. **Dataset preprocessing:** As there is one categorical variable(locations), I am using one hot encoding to convert into numerical variable of binary values(vectors)
2. **Normalizing the data:** the remaining numerical features have different ranges that are not the same ranges, I have normalized them using Z score method, such they all have same scale or range.
3. Now created the feature matrix with those and added the bias terms to the matrix. Then I have initialized the parameters learning rate, epochs, theta as array of zeros as given in the homework.
4. Then defined the code loop for the gradient decent and implemented or trained it on the feature matrix to get the theta values.
5. **Graphs:** Now evaluated the model by plotting the graphs of MSE loss curve over iterations. And actual v/s predicted curve too. Which came out to be as below.



- 5.1. **MSE Loss curve over Iterations:** As per the above graph, which is a downward slopping graph, I can say that the loss is decreasing as the no. of epochs is increasing and updating the parameters, which indicates that the model is converging to minimum error.
- 5.2. **Actual vs Predicted:** As per above graph I can say that model predicted the house prices accurately possible values as all predicted prices are on the line or along the line.
6. **Prediction:** After this I have predicted the house price for the given features
 - 6.1. Location: Downtown
 - 6.2. Size: 1900 sq ft
 - 6.3. Bedrooms: 3
 - 6.4. Bathrooms: 2
 - 6.5. Garage: Yes (1)
 - 6.6. Age: 5 yearsFor which the price of the house came to be **\$425,005.21.**

7. Discussion Questions:

Q1. FINAL LEARNING PARAMETERS(θ)

- A.** The **final learned parameters**(θ) are weights that are continuously updated and optimized through training of the gradient decent. They generally define the relationship between the features and target variable. These parameters will include terms like the bias and the coefficients of each feature in the linear equation.

They came to be(θ): [261733.23771591, 58980.18794367, 109431.38036405, 93321.66940819, 10757.96253002, -20706.88811296, -9973.28272017, 14902.84570662, -25558.52070766]

Q2. LEARNING RATE

- A.** i.) The Learning rate controls the size of each step taken in the opposite direction of the gradient during the gradient decent process.
- ii.) If it is high there are chances that the model may not converge or even diverge from the solution. Sometimes it may even overshoot the minimum.
- iii.) If it is low then the model makes small steps gradually but will eventually lead to slow convergence. If it is too low then the model will take a large to reach the convergence and optimal solution.
- iv.) So, we need the balanced learning rate that will help the model converge efficiently and effectively to reach optimal solution without overshooting. Given learning of 0.01 is good starting point that is effective for the model.